

Sharing Facial Cues at Different Target Positions to Support Virtual Collaboration

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Abstract

Virtual Reality (VR) collaboration allows users to share their emotions through visual cues placed at different locations. Various techniques have been proposed to enable VR collaborators to share their feelings with each other. However, it remains unclear how the placement of these cues affects collaboration in VR. This study explores how the location of emotional cues during a virtual collaboration task impacts users' attention, social interaction, and emotional experience. We created a VR prototype where users received their partner's emotions (neutral, happy, angry, sad and surprised) through facial expressions as emoji cues attached to different places: either their own hand, their partner's hand, or objects in the task environment. Testing these in an urban planning task, we found that the cues placed on a user's own hand attracted their partner's gaze, which improved social closeness and emotional understanding between the partners.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; *Collaborative interaction*.

Keywords

Virtual Reality, Collaboration, Social Computing, Visual Cue, Emotion, Emoji, Facial Expression

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1 Introduction

Sharing facial expression cues is becoming essential for communication in Virtual Reality (VR) collaboration. Previous research has introduced various ways for VR users to express their thoughts and emotions, such as virtual avatars [31] and visual props [6]. These visual representations are often shared with collaborators by attaching them to various places, such as their point of view [14, 36], objects [15, 19], or the surrounding environment [7, 16]. This demonstrates the flexibility of VR in connecting information to different contexts.

The dynamic placement of facial expression cues enables users to express themselves effectively, regardless of the task. For example, aligning facial expression cues with another user's perspective by displaying them on a screen can enhance the sense of presence and improve communication among VR collaborators [11]. Similarly, presenting facial expressions as a floating window in a VR environment has been shown to increase social presence and influence gaze behavior [35]. These studies highlight the benefits of placing facial cues in locations other than the avatar's face for improved

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visibility. However, it remains unclear whether their effectiveness stems from the intrinsic properties of the cues themselves or the specific locations where they are placed, as various confounding factors in the study design may influence the results. This is particularly relevant in collaborative situations where collaborators cannot directly view each other's faces (e.g., when working side by side). In such cases, exploring the placement of facial expression cues in more convenient locations could help maintain the benefits of facial expression cue accessibility, even when direct face-to-face interaction is not possible.

This research examines how the placement of facial expression cues influences VR collaboration. While previous research has explored the benefits of attaching cues to different targets, they often lack comprehensive testing conditions [11, 19, 35, 37]. To address these gaps, we isolated the impact of cue placement. We developed a collaborative VR system where two users could interact through virtual avatars in VR. To test the effects of cue placement, we introduced facial expression cues using emojis, attaching them to various targets such as the hands of the user or those of a collaborator, or on task objects. Facial expression cues were selected for their universal ability to convey emotions and foster social connections, which are critical for effective VR interaction [2, 5, 13, 18, 21, 23, 26, 28, 33]. The target locations for cue placement were chosen based on previous findings that suggested visual cues presented near a collaborator's gaze can enhance interaction [17, 18]. An urban planning task was designed to evaluate the effects of visual cues on collaboration in five conditions: (1) when no emojis are present, (2-3) when emojis are attached to either collaborator's hands (self-hand vs. collaborator's hand), and (4-5) when emojis are attached to a task object interacted with by either collaborator (self-object vs. collaborator's object). These attachment positions were selected based on the assumed locations where collaborators would primarily focus when solving the task. We found that visual cues placed on the user's own hand increased attention and improved social closeness and emotional understanding between collaborators. We formulated our research questions as follows: **RQ1**) How do different target positions affect users' attention in VR collaboration?; **RQ2**) How do different target positions influence users' social experiences in VR collaboration?; **RQ3**) How do different target positions impact users' affective experiences in VR collaboration?; and **RQ4**) What is the most suitable target position for facial expression cues to enhance VR collaboration?

2 Related Work

Sharing visual cues has been explored in various immersive collaboration scenarios, such as sharing natural gestures [36], facial expressions [21], or ray pointers [20] to convey thoughts and emotions during collaboration. For example, Teo et al. [36] introduced a technique for effective two-person collaboration by sharing hand gestures and visual annotations. In their approach, shared visual cues are attached to the collaborator's perspective. Their technique emphasizes sharing gestures in a first-person view to provide clearer instructions with fewer errors. It also creates the impression that both users are visually linked. Similarly, Hirsch et al. [12] proposed an asynchronous technique where a user's heartbeat is visualised as a trace-like pattern in a VR environment. Other users can revisit the

scene later and observe the user's actions and heartbeat, and feel socially connected. This demonstrates how sharing visual cues can be designed and used to offer distinct benefits. However, these methods also introduce challenges in assessing the advantages of visual cue sharing. Specifically, it becomes difficult to determine whether a technique's effectiveness comes from the visual representation itself or other factors, such as the position of the cues.

Placing visual cues at convenient locations, such as closer to the user [12, 39], has been shown to improve empathy [6, 8] and emotional understanding [34, 39]. Kim et al. [19] found that attaching gestures to task objects increased task efficiency and reduced mental effort. Similarly, Teo et al. [37] discovered benefits related to embodiment when arms were attached to task objects. However, these studies primarily focused on conventional communication cues, such as hand gestures, leaving a gap in research on other cues, such as emotion cues that are also important in collaboration.

Sharing emotions plays a crucial role in fostering empathetic collaboration. Facial expressions, in particular, are a common way to convey emotions [13] and build empathy [25]. Previous studies [18, 35] have investigated the placement of emotional cues and found benefits such as enhanced emotional interdependence [18], improved social presence [35], and better gaze behavior [35]. However, the proposed techniques were mostly limited to non-immersive environments or interactive task-specific settings.

Building on the gap in the existing literature, we explore how placing facial expression cues at different positions may influence VR collaboration. Our study focuses on convenient places for positioning visual cues, such as within the user's field of view, and close to where users are required to maintain their attention during collaboration. This includes potential target placements where users need to keep their focus such as users' hands or task objects.

3 System Overview

We developed a VR collaboration system using the Unity game engine (v2021.3.33f1) to test our research questions. The system allows two users to interact in a shared virtual environment as realistic avatars using the Meta Quest Pro Head Mounted Display (HMD). The HMD features a resolution of 1,800 x 1,920 pixels per eye, a 106° field of view, a 90Hz refresh rate, and trackers for facial expressions, head, eye and hand gestures. The Meta Movement SDK (v2.1.0) [29] was used to enable natural navigation and object interactions through gestural movements. The SDK also detects facial expressions by tracking muscle movements across 70 facial parameters. The avatars were designed with pale and achromatic skin tones to maintain consistency when measuring baseline experience data.

The system runs on HMDs connected to Windows 10 PCs (Intel Core i7-10870H CPUs, 16GB DDR4 RAM, and NVIDIA GeForce RTX 3060 GPUs) via a wired Quest Link. The Photon Pun 2 SDK [30] was used to synchronise interactions across the network. Data on avatar and object positions, rotations, and gestures (including facial, eye, and body tracking) are shared between users. For clarity, the user controlling their avatar is referred to as the **local** user, and their collaborator is the **partner** user.

For emotion sharing, facial expressions are converted into emojis, a common form of visual communication cue used in both research

and everyday life [1, 22, 24]. We selected five discrete emotions: Happy, Neutral, Sad, Angry and Surprise, as they are relevant across different ages and genders [4]. The system detects facial expressions by checking if their facial tracking parameters fall within predefined ranges, and generates the corresponding emojis. These parameters are mapped to the emojis based on standard muscle activity proposed by Matsumoto et al. [27], and colours are added to emojis according to Plutchik's Wheel of Emotions [32] for better visibility. Emojis are animated using Unity's particle system [38], where they appear and float away from a designated position. Following the user's own actions or their partner's actions, emojis are displayed either on the hands (in hand conditions) or on objects (in object conditions) (see Figure 1).

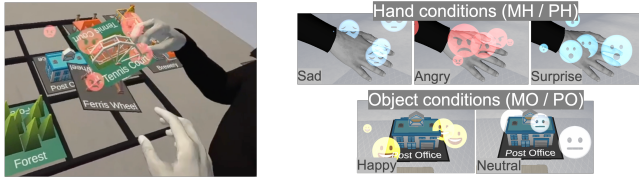


Figure 1: Facial expression cues and their placements. (Left) An example where the partner's emotions are shown as facial emojis on the object they interact with (a.k.a. PO condition). (Right) Five distinct emotions as different facial emojis.

4 User Study

We conducted a user study to assess how different placements of facial expression cues affect VR collaboration. The study included five conditions depending on where the partner's facial expression cues are presented to the local user: BL (Baseline without cues), MH (My Hand, i.e. a local user sees partner's facial cue on their own hand), MO (My Object, i.e. a local user sees partner's facial cue on the object which the local user interacts with), PH (Partner's Hand), and PO (Partner's Object). Users cannot see their own facial expressions in any of the conditions. These positions were chosen based on the assumption that users would primarily focus on either the hand or task object, as the task involves interacting with objects using the hands. We expect that placing facial expression cues closer to the local user will make it more convenient for them to see their partner's emotions, while placing them on the partner will make it more intuitive to connect the emotions directly to the partner. The study adopted a within-subject design and examined how these varying positions of facial expression cues influence users' attention, social connection, and social relationships during a synchronous collaboration task in VR.

4.1 Task Environment and Design

The experiment was conducted in a 3x3 meters room divided by a partition. Each side of the partition was equipped with a chair, a desk, and a computer. In the VR environment, a table was placed at the centre of the room with three distinct areas (see Figure 2). There were nine objects in the "task pick" area. Participants were required to read the task instructions in the "task instruction" area, and collaborate to place all nine objects in the correct 3x3 grid positions

in the "task put" area to complete the task. The instructions involved directional terms, such as "The Forest is NORTH of the City Hall", "The Pizzeria is NEXT to the Tennis Court", and "The Brewery is at a CORNER".

The task required two participants to work together while standing side by side in front of a table, communicating task instructions and moving objects to their correct locations. To ensure cooperation while simulating a scenario where understanding emotions is beneficial for problem-solving, each participant was given a different subset of the task instructions. There were five unique pairs of task instruction subsets to ensure that the tasks did not repeat across one training session and four task conditions. It was not possible for a single participant to complete the task alone. During the task, a red button appeared every 20-25 seconds at one of nine predefined positions in the "task button" area of the table. When the button appeared, the task objects were hidden, and the participant needed to click the button to make the objects reappear. The button functioned independently for each participant, meaning each user had to click their own button to resume the task. This button acted as a minor interruption (e.g., receiving a text message) to prolong the task without significantly increasing its difficulty. It also served as a controlled distraction, designed to elicit different emotional responses from participants as they reacted to these unexpected disruptions.

A progress indicator showed the task's progress. It displayed four stages: "<25%", "<50%", "<70%", and "ALMOST". It only appeared when both participants showed matching facial expressions, excluding neutral. This feature emphasized moments of emotional consensus, encouraging participants to pay more attention to each other's emotions. It mirrors real-life scenarios where successful collaboration relies on emotional and cognitive harmony, rewarding participants for achieving emotional alignment and building mutual emotional understanding. The task design was inspired by an urban planning task from [31] where participants collaborated to solve the problem together.

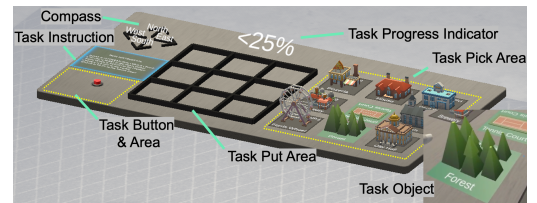


Figure 2: Urban planner task overview.

4.2 Study Procedure

The procedure started with participants signing a consent form, reviewing instructional slides, undergoing eye calibration, and putting on the HMD at the centre of the task area. Then, a training session began by immersing participants in the task environment where they completed the urban planning task without facial expression cues. During the training session, participants were also shown an instruction slide with five pictures representing different facial expressions. They practiced how to make these expressions, and a text displayed their currently detected emotion. The goal was to

help participants become familiar with the task procedure and the system features.

After the training session, participants removed the HMD and answered pre-experiment demographic questions before proceeding to the main task. The main task was divided into five sessions, each corresponding to a different condition. In each session, participants completed the urban planning task under the assigned condition. To begin, they were instructed to stand in front of the table and adjust their position for comfortable interaction with the objects. Once the investigator confirmed the participant's position and visibility of the task instructions and objects, they clicked a button to start the task. After completing the task, participants were helped to remove the HMD and answered a post-task questionnaire. They then repeated a similar procedure for the subsequent sessions. The order of conditions and task instructions was counterbalanced using a Balanced Latin Square design. However, since the number of task sessions ($N=13$) was not a multiple of the number of conditions ($N=5$), the first five unique orders generated by the Balanced Latin Square were repeated to accommodate all participants. The experiment concluded after participants completed a post-experiment questionnaire. The whole procedure took about 90 minutes.

5 Results

We recruited 26 participants (13 females, 13 males) in pairs. The majority of participants (39.3%) were aged 26-30, followed by those aged 18-25 (28.6%), 31-35 (17.9%), 36-40 (7.1%), and 41-50 (7.2%). Most participants reported themselves were not familiar with VR: 21.4% had never used it, 32.1% used it a few times a year and 17.9% used it a few times a month on a multiple-choice question. Additionally, 21.4% were weekly users and 7.1% used VR daily. Most participant pairs (85.7%) knew each other. Regarding experience with logic puzzles or games based on a custom question "Rate your experience on logic puzzles task or game (E.g. Sudoku)" on a 7-point Likert scale (1:extremely inexperienced - 7: extremely experienced), 2 participants reported being inexperienced, 19 had some experience and 5 considered themselves very experienced. We report the results with statistical analyses ($\alpha = .05$, unless noted otherwise). The Shapiro-Wilk test indicated that variables are not in a normal distribution hence non-parametric tests are conducted.

We measured participants' gaze behavior as an objective metric. We also gathered subjective feedback on their experiences in each condition through several questionnaires: the Networked Minds Measure of Social Presence (NMMSP) [9], emotional responses using the SAM [3], and the Inclusion of Other in Self (IOS) [41]. All of these were included in the post-task questionnaire. At the end of the study, participants answered their preferences and reasons for them through subjective comments in the post-experiment questionnaire.

5.1 Gaze Activity

Figure 3 shows the participants' average gaze duration on different objects in the task environment, normalized to 100%. The objects were categorized into four main groups: the Local's avatar (MyAvatar), the Partner's avatar (PartnerAvatar), task objects (TaskObject) and miscellaneous objects such as the table, wall, and floor (ETC). To avoid subjective assumptions in describing gaze activity,

the time spent gazing at facial expression cues was categorized according to their attachment. For instance, MyAvatar was considered in the MH condition, TaskObject in the MO and PO conditions, and PartnerAvatar in the PH condition. The Friedman test indicated a significant difference for MyAvatar ($\chi^2(4)=39.1876$, $p<.0001$). A posthoc analysis using the Wilcoxon Signed Rank test with Bonferroni correction ($\alpha = .01$) revealed significant differences between the MH condition ($M = 14.72$, $SD = 6.5460$) and all other conditions: BL ($M = 5.40$, $SD = 4.8976$, $Z=0.0889$, $p<.0001$), MO ($M = 5.12$, $SD = 3.8485$, $Z=0$, $p<.0001$), PH ($M = 8.12$, $SD = 14.2844$, $Z=0.4191$, $p=.0001$), and PO ($M = 6.31$, $SD = 6.2554$, $Z=0.1651$, $p<.0001$). Additionally, the Friedman test showed a significant difference for the PartnerAvatar category ($\chi^2(4)=16.2154$, $p=.0027$), and posthoc analysis revealed a significant difference ($Z=0.8127$, $p=.0036$) between the MH ($M = 12.08$, $SD = 8.0865$) and PH conditions ($M = 18.14$, $SD = 10.6539$). The Friedman test also found a significant difference for the TaskObject category ($\chi^2(4)=16.7077$, $p=.0022$), with a posthoc test showing a significant difference ($Z=0.8508$, $p=.0047$) between the MO ($M = 31.98$, $SD = 12.7205$) and PH conditions ($M = 24.15$, $SD = 11.9875$). No significant difference was found for the ETC category.

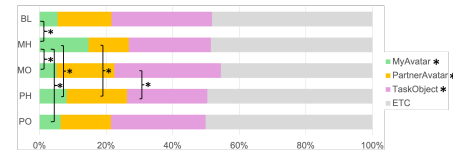


Figure 3: Participants' gaze behavior on different categories by percentage (*=Statistically Significant).

5.2 Social Presence

To evaluate participants' social presence across conditions we analyzed the sub-scales of Co-Presence (CP), Attention Allocation (AA), Perceived Message Understanding (PMU), Perceived Affective Understanding (PAU), Perceived Emotional Interdependence (PEI), and Perceived behavioral Interdependence (PBI), as well as the overall scale (SP). The NMMSP [9] included 36 rating items on a 7-point Likert scale (1: strongly disagree - 7: strongly agree). We analyzed the results as a whole and for each sub-scale. However, no significant differences were found. We then compared each sub-scale against the medium score ($N=4$). Interestingly, PEI showed significantly higher-than-average ratings in the MH ($M = 4.86$, $SD = 1.4015$, $Z = 0.6349$, $p = .0074$), MO ($M = 4.85$, $SD = 1.4053$, $Z = 0.7175$, $p = .0074$), PH ($M = 4.54$, $SD = 1.3757$, $Z = 0.8826$, $p = .0369$), and PO ($M = 4.65$, $SD = 1.2466$, $Z = 0.7746$, $p = .0190$) conditions, but not in BL. Results where no significant differences were observed between conditions were excluded, as they were considered equivalent to the BL condition.

5.3 Inclusion Of Other in the Self

We analysed the single IOS scale across different conditions. The IOS scale [41] used a 7-point Likert scale (1: no overlap in relationship - 7: most overlap). The Repeated Measures ANOVA test with Aligned Rank Transform [40] showed no significant results across

conditions. However, when compared to the mean value ($M = 4$), we found that the feeling of connection with others was significantly higher in the MH condition ($Z = 0.7619$, $p = .0275$).

5.4 Preferences

Participants rated their preference for each condition on a scale from least (1) to most (5) preferred, with one condition per ranking category. Overall, the preferences were split among the conditions. The highest number of participants ($N=7$, 26.9%) rated BL as their most preferred condition, followed by the MH condition ($N=6$, 23.1%). Interestingly, the highest number of participants ($N=10$, 38.5%) also rated BL as their least preferred condition, with the PH condition ($N=9$, 34.6%) coming next. A notable number of participants ($N=10$, 38.5%) rated MH as their second-most preferred condition. Despite these observations, no significant differences were found using the ART-ANOVA test.

When participants were asked about what they liked or disliked in each condition, the majority in the BL condition ($N=18$) appreciated the collaborative nature of the VR experience. However, it was unclear whether this enjoyment stemmed from the experience itself or from the absence of facial expression cues. Six participants expressed dissatisfaction with not being able to see their partner's emotions, which created a sense of emotional distance. P25 said, *"without facial visual feedback, I felt emotionally distant from my partner"*. Even though users could view their partner's face by looking directly at their avatar, they still felt emotionally disconnected. This suggests that the limitations of the condition led participants to split their attention between their partner's avatar and task objects.

In the MH condition, six participants found the emojis helpful for understanding their partner's emotions and improving interaction. P11 said, *"I was able to tell how my partner felt through the emoji"*. On the other hand, five participants found the emojis either distracting or difficult to notice. For instance, P5 commented, *"I didn't notice my partner's emotion on my hand most of the time, maybe because I was too focused on the task"*. P25 added, *"I felt strongly emotionally connected to my partner in this task, though it wasn't helpful for solving the task"*, suggesting that while the emojis enhanced emotional understanding, they did not always contribute to task performance.

In the MO condition, seven participants noted it was easier to interpret their partner's emotions. P3 stated, *"I like the idea of the emotion on my object so that I can see my partner's emotion more easily"*, and P24 added, *"I can see my partner's emojis better this time"*. However, four participants found the emojis distracting when not needed. Both P3 and P7 mentioned that the movement of emojis and *"icons flying away"* were distracting.

In the PH condition, five participants found it easy to track their partner's emotions through the facial expression cues on their hand. P3 commented, *"I can follow my partner's emotions easily thanks to the emojis on their hand"*. However, four participants found the emojis less noticeable, especially when their partner's interactions were minimal. P5 noted, *"the emotion display was not noticeable because my partner didn't place the items much"*. Additionally, six participants highlighted difficulties in seeing emotions due to avatar overlap, position of the cue, and object occlusion.

In the PO condition, five participants appreciated the emoji placement for understanding their partner's feelings. P23 said, *"I can*

see the emojis of my partner, so I understand how the task is going" and P24 added, *"emojis on objects are easier to spot compared to emojis on the partner's hand"*. P5 participant also mentioned, *"I was sometimes influenced by my partner's moods; when my partner laughed, I laughed with him"*. However, four participants found the emojis distracting or confusing, especially when the ownership of objects changed. As P25 remarked, *"it is a little confusing when emojis change on an object based on who is holding it"*.

6 Discussion and Limitation

Regarding RQ1 (How do different target positions affect users' attention in VR collaboration?), our findings suggest that adding facial expression cues to the hand of the local or partner user generally improves attention towards a partner's emotions. Participants frequently looked at their own hand in the MH condition, effectively focusing more on the facial expression cue. In contrast, we observed no significant changes in behavior in the PO condition. We believe that participants primarily focused on their own task, which led them to direct attention mainly to their own hand or the object they were interacting with. However, it is important to note that attention towards the partner's avatar was greater in the PH condition compared to the MH condition. This may be due to the increased attention paid to their own avatar in the MH condition. In other words, the MH condition showed the most attention towards the user's own avatar across all conditions, which led to a significant difference between the MH and PH conditions.

Regarding RQ2 (How do different target positions influence users' social experiences in VR collaboration?), we found no evidence of the effects when comparing different positions of facial expression cues. However, all conditions, except for BL, showed above-average perceived emotional interdependence. This suggests that having access to facial expression cues when needed, at convenient locations, is sufficient to enhance the understanding of a partner's emotions.

Regarding RQ3 (How do different target positions impact users' affective experiences in VR collaboration?), previous research acknowledges that placing facial expression cues near the gaze point increases emotional interdependence [17]. Our results on the PEI subscale of Social Presence support this. Moreover, regarding IOS results, only the MH condition reported an above-average IOS score. We believe this may be due to the facial expression cue being attached to the user's body parts, which likely triggered an embodiment experience, making the partner's emotions feel more integrated with their own. Additionally, the higher attention directed towards their own hand in the MH condition, compared to the other conditions, may have contributed to the higher IOS score.

Regarding RQ4 (What is the most suitable target position for facial expression cues to enhance VR collaboration), there was no clear "winning" condition. However, participants' comments provided various perspectives on the pros and cons of each condition. Those favouring the BL condition suggested that the convenience of having facial expression cues depends on the task requirements. In this case, participants felt that understanding their partner's emotions was less relevant for completing the task despite the benefits of seeing their task progressions when having similar emotions. While we expected participants to prefer the MH or MO conditions due to their high accessibility, some participants also found these

positions distracting. On the other hand, placing the cues on their partner's hand or object made them less obtrusive, although the accessibility then depended on the partner's interactivity. This indicates that the relevance and necessity of the facial expression cues are more important than their accessibility. Although attaching facial expression cues to the local participant's own hand in the MH condition increased attention to their partner's reactions and resulted in above-average IOS scores, it did not necessarily make it the most preferred position for observing their partner's emotions.

Overall, the placement of facial expression cues had varied effects. Attaching them to the local's hand captured attention and enhanced social experiences but was also distracting. Placing cues on the partner's hand was beneficial depending on the partner's engagement. A study limitation was the predefined emoji parameters, which affected emotion recognition. Neutral, Happy, and Surprise were detected more easily, while Angry and Sad had lower sensitivity, making some emotions appear more frequently and requiring users to express themselves more consciously. However, since users couldn't see their own emojis, this likely had minimal impact as a confounding factor. Regardless, improving emotion detection could enhance cue effectiveness.

The task design could be improved to make emotions more relevant and useful. While each participant pair received a unique task subset, the task provided little motivation for varied facial expressions. The five emojis were chosen for consistent interpretation across ages and genders, but not all were equally useful. Sad and Angry were likely less relevant than Neutral, Happy, and Surprise. Since our focus was on how facial expression cue placement affects collaboration rather than analysing the cues themselves, this had minimal impact on our findings. However, future research could design tasks that naturally elicit a wider range of emotions to explore their role in collaboration. Additional measures like task load and physiological responses could offer deeper insights into cue placement's psychological effects. Although we measured factors like task completion time, eye blinks, and facial activity, no significant differences emerged. Future studies could refine the experimental design to explore different task contexts. We also did not control for participants' prior relationships, which might have influenced emotional understanding, though this was likely minor due to our within-subject design with counterbalanced conditions. Future research will address these limitations to enhance the understanding and application of facial expression cue placement in VR collaboration.

7 Conclusion and Future Work

We studied how different placements of facial expression cues affect VR collaboration using an urban-planning task. Participants saw their partner's emotions as facial emojis while working together in VR. We compared five conditions: no cue (baseline) and four placements on the user's or their partner's hand or objects. We found placing the partner's facial expression cue on the participant's hand increased attention, improving emotional understanding and enhancing feelings of closeness and interdependence. However, no clear preference for a particular placement was found. Future research will explore different VR collaboration setups and cue placements, such as aligning them with gaze or head movements to

keep them in view. While past research noted visibility issues with avatar heads [10], emoji representations may improve this. These refinements will deepen our understanding of cue placement's role in VR collaboration.

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